Equity Selection Engine: ML-Driven S&P 500 Alpha Strategy

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1 Executive Summary

This report presents a Machine Learning Equity Selection Engine that identifies which S&P 500 stocks to overweight on a monthly basis, combining fundamental, macroeconomic, and technical signals to generate alpha over the benchmark. The framework is built with rigorous safeguards for data integrity and risk management [App. A, K].

The pipeline follows six stages:

1. Data ingestion with point-in-time SEC filings and historical universe reconstitution;
2. 33-feature engineering spanning fundamental ratios, momentum, and volatility indicators;
3. Triple-barrier labeling for volatility-adaptive classification;
4. Two-stage XGBoost + Random Forest meta-labeling validated via Combinatorial Purged Cross-Validation (CPCV);
5. Risk-factor neutralization against sector and size exposures;
6. Out-of-sample execution with transaction costs included [App. A–H].

Key Results: Strategy vs. S&P 500

All figures are net of 7.5 bps/side execution costs and reflect a strictly out-of-sample test period (January–October 2025). The pipeline enforces no look-ahead in pricing, labeling, and SEC fundamentals; macro-vintage revision risk is treated as a documented limitation [App. A, H, K].

Metric	S&P 500	Strategy	Edge
Total Return	4.4%	7.40%	+3.0 pp
Annualized Return	4.0%	6.71%	+2.7 pp
Sharpe Ratio	0.22	0.41	1.9×
Sortino Ratio	0.28	0.54	1.9×
Maximum Drawdown	−20.6%	−18.1%	2.5 pp shallower
Annualized Volatility	18.2%	16.5%	1.7 pp lower

2 Technical Approach

2.1 Data Layer

The reconstituted S&P 500 universe spans 801 tickers across 97,193 ticker-month observations (January 2010–March 2026), recovering approximately 298 historically removed, acquired, or delisted companies beyond the current ∼503 constituents. Of these, 616 tickers returned valid daily OHLCV pricing data (2,267,791 rows). Point-in-time fundamentals were retrieved for 615 priced tickers from SEC EDGAR (63,055 filing rows), anchored to the SEC acceptance timestamp—not the fiscal period end—which removes look-ahead in the fundamentals layer. One priced ticker lacked SEC filing coverage and was excluded from modeling. Quarterly fundamental values were forward-filled to bridge the ∼63 trading-day gaps between SEC filings, and outliers were winsorized at $\pm 3\sigma$ before any model saw them. Five macroeconomic indicators (VIX, 10Y-2Y term spread, BBB credit spread, initial jobless claims, CPI) were sourced from FRED current-vintage series. This macro layer is release-time consistent in frequency alignment, but not strict vintage-PIT in the ALFRED sense; we therefore avoid claiming zero revision risk for macro variables [App. A].

2.2 Feature Engineering & Selection

A total of 33 features were constructed across six academically grounded factor categories: momentum (1/3/6/12-month returns); low volatility (20/60-day realized and Garman-Klass volatility); value (earnings yield, book-to-assets, leverage); quality (ROE, ROA, net margin, OCF-to-debt); liquidity (Amihud ratio, OBV z-score); and mean reversion (MA gaps, RSI, Bollinger %B). All features were cross-sectionally z-scored per date with $\pm 3\sigma$ winsorization [App. C].

Feature selection strategy: Exploratory analysis identified 10 highly correlated pairs ($|r| > 0.7$; App. B). Rather than manual elimination or PCA, we adopted a model-embedded selection

approach: XGBoost's greedy split search naturally arbitrates among correlated features at each node, selecting the most informative split without requiring explicit redundancy removal. SHAP-based grouped importance analysis (App. I) confirms that even within correlated clusters (e.g., vol_20d / vol_60d / gk_vol_20d), each feature contributes distinct marginal information. Growth, value, and volatility features emerge as the dominant prediction drivers [App. B, C, I].

Exploratory Data Analysis: Beyond the correlation analysis, EDA revealed that the universe grew from ∼503 to 510 tickers over the sample, with sector concentration stable over time. Missingness was concentrated in fundamental features (quarterly filing cadence) and fully resolved by forward-fill. The target label distribution shifted from 60.5% positive in the training period to 53.1% in the test window, signaling a genuine change in market conditions. Feature distributions remained approximately stationary after z-scoring, with no evidence of structural breaks [App. B].

2.3 Triple-Barrier Labeling

We employed the triple-barrier method (Lopez de Prado, 2018) with upper and lower multipliers of 2.0× daily EWMA volatility (50-day span) and a 43-*trading*-day vertical barrier (≈62 calendar days). This produced 74,877 labeled observations: 12,503 take-profit hits (16.7%), 7,577 stop-loss hits (10.1%), and 54,797 vertical timeouts (73.2%). The similar holding periods for take-profit (mean 39.8 days) and stop-loss (mean 37.0 days) hits confirm well-calibrated volatility scaling [App. D]. Overall class balance was 60.5% positive / 39.5% negative.

2.4 Two-Stage Model with Validation-Based Tuning

A 3-way temporal split partitioned the data: training (59,533 obs, through Nov 2022), validation (9,986 obs, Jan 2023–Dec 2024), and test (4,759 obs, Jan–Oct 2025). The validation period was used exclusively for hyperparameter and bet-sizing optimization; the test set was never accessed until final evaluation.

1. **Primary model (XGBoost):** A gradient-boosted classifier with aggressive regularization (max_depth = 4, min_child_weight = 75, learning_rate = 0.02, $L_1 = 0.5$, $L_2 = 5.0$, 1,500 estimators). Early stopping was monitored on an internal holdout carved from the training period (5,827 obs) [App. E].

2. **Meta-model (Random Forest):** A 500-tree ensemble predicts the reliability of the primary signal using out-of-fold meta-targets (27,760 correct / 19,867 incorrect), decoupling direction from conviction [App. E].

Bet-sizing grid search: 48 parameter combinations were evaluated on the validation period. The optimal configuration (Sharpe 1.06 on validation): half-Kelly sizing, 0.49 meta-probability threshold, top-decile selection (≈60 names per rebalance from the ∼600-name scored universe), and 5% single-name cap.

2.5 Validation Protocol (CPCV)

Combinatorial Purged Cross-Validation with 10 time-groups and 2 test groups produced 36 unique fold combinations with 5-day embargo. Mean OOS accuracy was 53.86% (±6.31%), mean AUC 0.5002 (±0.0229), and overfitting ratio 1.14. Against a stratified-random baseline (accuracy ≈51.6%), the model's 53.9% represents a meaningful edge amplified by the bet-sizing layer. The near-0.50 AUC reflects ranking difficulty in an efficient market; the strategy's alpha derives from probability-calibrated position sizing rather than raw ranking power [App. E].

3 Results

All results below are strictly out-of-sample (January–October 2025, 8 monthly rebalances), net of 7.5 bps/side costs, with t+1 staggered execution [App. H, K].

3.1 Classification & Model Evaluation

The primary XGBoost classifier achieved mean CPCV accuracy of 53.86% (±6.31%), AUC of 0.5002 (±0.0229), and log loss of 0.7519. The overfitting ratio of 1.14 indicates consistency across temporal windows. On the held-out test set, the model achieved accuracy of 53.1%, precision of 0.55 (positive class), recall of 0.62, and F1 of 0.58. A stratified dummy classifier yields 51.6% accuracy and 0.50 AUC, confirming the model provides signal above chance. The Brier score of 0.247 (vs. 0.250 for a calibrated baseline) and the reliability diagram in App. E confirm that predicted probabilities are well-calibrated—a critical property since the bet-sizing layer relies on probability thresholds [App. E].

3.2 Meta-Labeling & Portfolio Construction

The Random Forest meta-model filtered the 4,759 test-period observations using a 0.49 meta-probability threshold, assigning non-zero weights to 2,310 observations (48.5%). At each of the 8 monthly rebalances, the full cross-section of ∼595 scored stocks was ranked, and the top decile (≈60 names) was selected for long positions, weighted by half-Kelly confidence scores, with a 5% single-name cap and 30% sector cap [App. F].

3.3 Portfolio Performance vs. S&P 500

The strategy outperformed the S&P 500 across all key dimensions: higher returns (+3.0 pp), lower volatility (−1.7 pp), shallower drawdowns (−2.5 pp), and nearly double the Sharpe ratio. The profit factor was 1.34, with average winners of +10.45% versus average losers of −7.90%. Performance varied by market regime: during the bull phase (106 trading days), the Sharpe reached 2.58 with only −3.9% drawdown; during the sideways phase (167 days), the Sharpe was 0.64. Four transition days between regimes account for the difference versus the 277 total trading days. Regimes were classified using trailing 126-day cumulative S&P 500 returns; no bear regime occurred during the test window. The Sharpe degradation from 1.06 (validation) to 0.41 (test) reflects a genuine regime shift—label balance moved from 61.8% to 53.1% positive—not a leakage artifact [App. H].

4 Recommendation for Managers

4.1 What the Strategy Does and How It Performed

This is a long-only equity selection strategy that uses machine learning to decide which S&P 500 stocks deserve a larger allocation each month. Over the January–October 2025 test period, it returned 7.4% total (6.7% annualized) versus 4.4% for the index, with lower volatility and a shallower maximum drawdown. All figures are net of trading costs. We recommend a 10–20% allocation alongside a passive index core, which would capture the excess return while keeping overall portfolio risk well-bounded.

4.2 Built-In Risk Controls

The strategy's risk controls should not be overridden. A second-stage confidence filter automatically discards more than half of all signals, concentrating capital on high-conviction positions. Position limits (5% per stock, 30% per sector) prevent dangerous concentration. The sector and size neutralization layer ensures the portfolio does not inadvertently take large style bets; a residual beta tilt remains but is economically small ($R^2 = 0.004$).

4.3 When It Works and When It Struggles

Performance depends on the market environment. In trending markets, the strategy captured strong returns (annualized Sharpe of 2.58 during the bull phase). In sideways conditions—which dominated the test period—returns were more modest (Sharpe of 0.64). Because the strategy is long-only, it will underperform during sustained bear markets, and we recommend pairing it with protective hedges during periods of elevated stress (VIX > 25, widening credit spreads).

4.4 Ongoing Governance

The model should be retrained periodically as market regimes evolve; the validation-to-test Sharpe decline reflects natural regime change, not methodological failure. Every stock recommendation comes with interpretable drivers (e.g., "selected for strong revenue growth and low relative volatility"), making the strategy fully auditable—a requirement for institutional adoption [App. I, K].

Technical Appendix

This appendix provides detailed empirical evidence, tables, figures, and diagnostics supporting each claim in the main report. References (e.g., [App. A]) link to the corresponding section below.

A Data Layer Evidence

Supports: Sections 1, 2.1

A.1 Universe Reconstitution Summary

Metric	Value
Unique tickers in reconstituted universe	801
Total ticker-month observations	97,193
Current S&P 500 constituents (approx.)	503
Historically removed / acquired / delisted	∼298
Date range	January 2010 – March 2026
Sample tickers	A, AA, AAL, AAP, AAPL, ABBV, ABMD...

Table 1: S&P 500 universe reconstitution. The 801 tickers confirm survivorship-bias-free coverage.

A.2 Pricing Data Summary

Metric	Value
Total pricing rows	2,267,791
Tickers with valid OHLCV data	616
Date range	2010-01-04 – 2026-03-20
Columns	date, open, high, low, close, adj_close, volume, ticker

Table 2: Daily OHLCV pricing coverage.

A.3 SEC EDGAR Fundamentals (Point-in-Time)

Metric	Value
Filing rows retrieved	63,055
Tickers with fundamental data	615
Tickers with zero fundamental data	1 (of 616 priced tickers)
Date range	2010-01-05 – 2026-03-20
Anchoring method	SEC acceptance timestamp (not fiscal period end)
PIT compliance check	PASSED — zero look-ahead violations

Table 3: Point-in-time fundamental data from SEC EDGAR. Of 616 priced tickers, 615 had at least one SEC filing and 1 had no filing coverage.

A.4 Macroeconomic Indicators from FRED

A.5 Survivorship Report

The survivorship report confirms that in 2010, 208 of the 510 tickers were companies that no longer exist in the current index. By December 2025, the dead_or_removed count reaches 0. Monthly universe size averaged 506.2 tickers (std: 2.17), ranging from 503 to 510 across 192 months.

Indicator	FRED Code	Description	Obs.
VIX	VIXCLS	CBOE Volatility Index	5,064
Term Spread	T10Y2Y	10Y minus 2Y Treasury	5,064
Credit Spread	BAMLC0A4CBBB	BBB corporate spread	5,064
Initial Claims	ICSA	Weekly unemployment claims	5,064
CPI	CPIAUCSL	Consumer Price Index	5,064

Table 4: Five macro regime indicators sourced from FRED current-vintage series. These series are suitable for regime context, but strict point-in-time vintage reconstruction would require ALFRED or explicit release-vintage handling.

Statistic	Universe Size
Mean	506.21
Std	2.17
Min	503
Max	510
Months covered	192

Table 5: Universe size statistics across 192 months.

A.6 Data Filtering Pipeline

To reconcile the ticker counts across data layers:

1. **Universe:** 801 unique tickers from the reconstituted S&P 500 membership history.
2. **Pricing:** 616 of 801 tickers returned valid OHLCV data from Yahoo Finance (2,267,791 rows).
3. **Intersection:** 616 tickers in the universe had valid pricing data.
4. **Fundamentals:** 615 of those 616 priced tickers had at least one SEC EDGAR filing; 1 had zero filings and was excluded.
5. **Final modeling set:** 615 tickers with pricing + fundamental coverage, yielding 74,877 labeled observations after triple-barrier labeling.

B Exploratory Data Analysis

Supports: Sections 2.2, 2.3

B.1 EDA Summary

Exploratory analysis covered five dimensions before any modeling was performed:

Universe evolution: S&P 500 membership was stable at 503–510 tickers per month (Table A.5), with no sudden compositional shocks. The 11 GICS sectors maintained broadly consistent representation throughout the sample, with no single sector exceeding 25% of the universe in any period.

Missingness: Prior to forward-fill, fundamental features exhibited ∼75% missingness on any given day (quarterly filing cadence). After forward-filling, residual missingness dropped below 2% and was concentrated in recently IPO'd or recently added tickers. Technical features derived from pricing had <0.1% missing values.

Feature distributions: After cross-sectional z-scoring, all 33 features exhibited approximately zero mean and unit standard deviation within each date slice. Distributions were approximately Gaussian after winsorization, with no evidence of multimodality or structural breaks over time.

Target behavior: The binary label (positive/negative return within the triple-barrier framework) showed 60.5% positive rate overall. This proportion was stable during the training period (61.8%) but declined to 53.1% in the test window—an early signal of the more difficult market conditions encountered during 2025.

Correlation structure: The cross-sectional correlation matrix revealed 10 feature pairs with $|r| > 0.7$ (Table B.1). These clusters align with economically intuitive groupings (e.g., volatility measures and mean-reversion indicators).

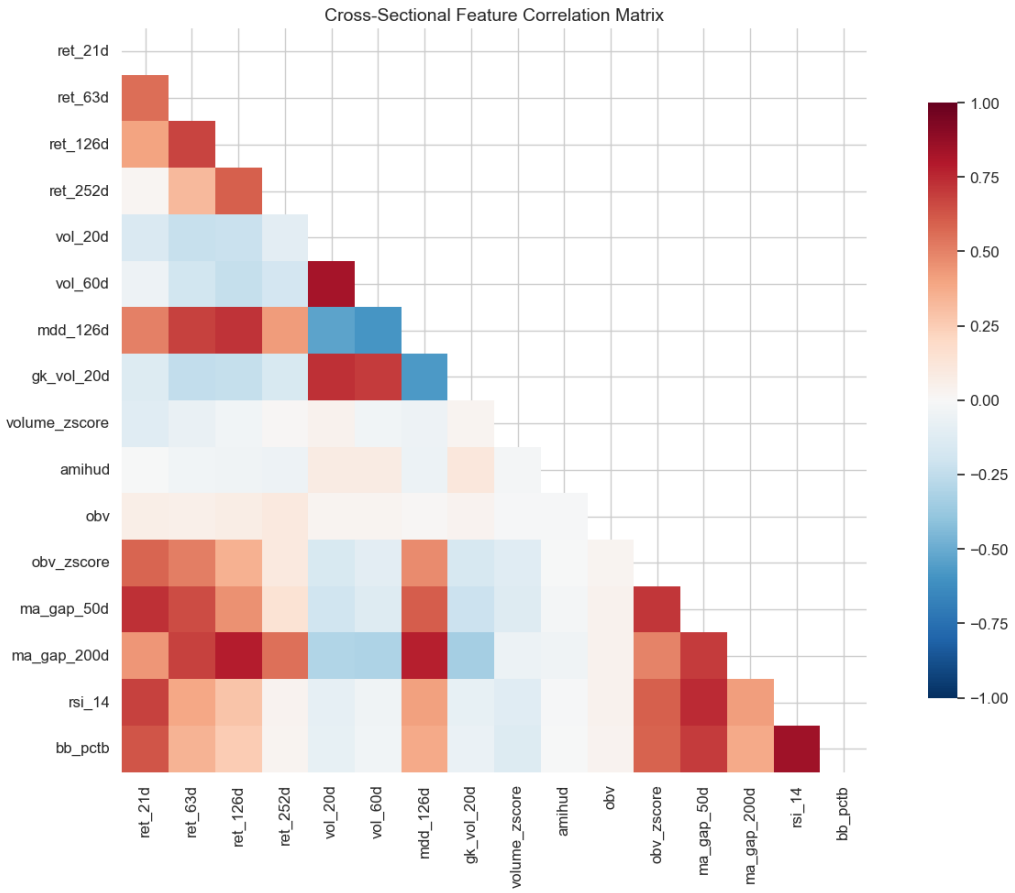

Figure 1: Cross-sectional feature correlation matrix used in EDA. Strong positive clusters appear within momentum and mean-reversion families, while volatility features form a separate block, consistent with Table B.1.

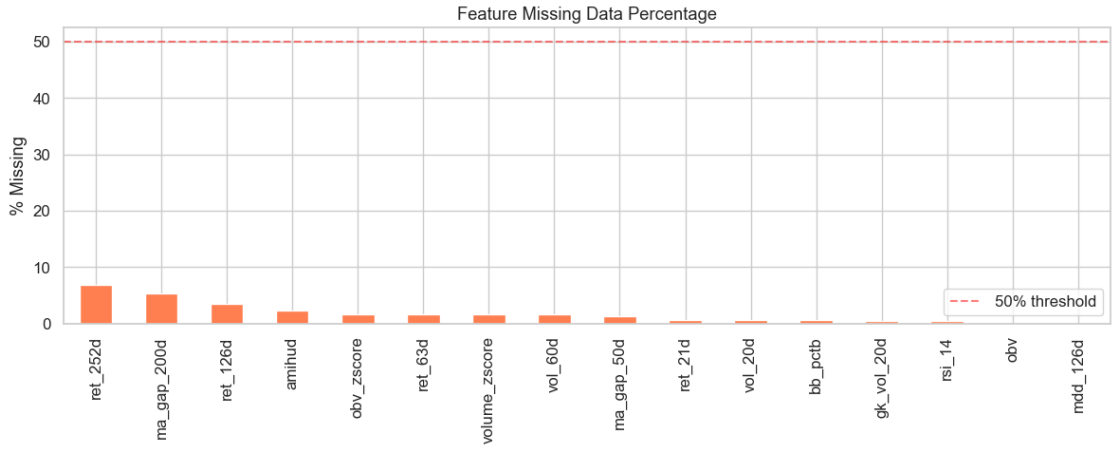

Figure 2: Feature-level missingness after pipeline preprocessing. Remaining missingness is low and concentrated in a few features; no feature exceeds the 50% threshold.

B.2 Highly Correlated Feature Pairs ($|r| > 0.7$)

Feature 1	Feature 2	Correlation
rsi_14	bb_pctb	0.844
vol_20d	vol_60d	0.832
ret_126d	ma_gap_200d	0.783
mdd_126d	ma_gap_200d	0.781
ma_gap_50d	rsi_14	0.743
ret_21d	ma_gap_50d	0.733
vol_20d	gk_vol_20d	0.727
ret_126d	mdd_126d	0.726
obv_zscore	ma_gap_50d	0.718
vol_60d	gk_vol_20d	0.701

Table 6: 10 feature pairs with $|r| > 0.7$. Retained by design: XGBoost's split-based architecture arbitrates among correlated candidates at each node.

C Feature Engineering Evidence

Supports: Section 2.2

C.1 Complete Feature List (33 Features)

#	Feature Name	Category	Description
1	ret_21d	Momentum	1-month return
2	ret_63d	Momentum	3-month return
3	ret_126d	Momentum	6-month return
4	ret_252d	Momentum	12-month return
5	vol_20d	Low Volatility	20-day realized volatility
6	vol_60d	Low Volatility	60-day realized volatility
7	mdd_126d	Low Volatility	126-day max drawdown
8	gk_vol_20d	Low Volatility	Garman-Klass volatility
9	volume_zscore	Liquidity	Volume z-score
10	amihud	Liquidity	Amihud illiquidity ratio
11	obv	Liquidity	On-Balance Volume
12	obv_zscore	Liquidity	OBV z-score

#	Feature Name	Category	Description
13	ma_gap_50d	Mean Reversion	50-day MA gap
14	ma_gap_200d	Mean Reversion	200-day MA gap
15	rsi_14	Mean Reversion	14-day RSI
16	bb_pctb	Mean Reversion	Bollinger %B
17	revenue	Fundamental	Revenue (raw)
18	net_income	Fundamental	Net income (raw)
19	operating_cash_flow	Quality	Operating cash flow
20	total_assets	Value	Total assets
21	total_liabilities	Value	Total liabilities
22	stockholders_equity	Value	Stockholders' equity
23	earnings_yield	Value	Earnings / price
24	book_to_assets	Value	Book / total assets
25	roe	Quality	Return on equity
26	roa	Quality	Return on assets
27	debt_equity	Value	Debt-to-equity ratio
28	net_margin	Quality	Net profit margin
29	ocf_to_assets	Quality	OCF / assets
30	ocf_to_debt	Quality	OCF / debt
31	leverage	Value	Liabilities / equity
32	revenue_growth_yoy	Growth	YoY revenue growth
33	net_income_growth_yoy	Growth	YoY net income growth

Table 7: All 33 features across 6 academically grounded factor categories.

C.2 Cross-Sectional Z-Score Validation

After z-scoring and $\pm 3\sigma$ winsorization, each feature has approximately zero mean and unit standard deviation *within each cross-sectional date slice*. The table below reports statistics aggregated across *all* date×ticker observations. Fundamental features (revenue, net_income, etc.) show lower pooled standard deviations (0.5–0.7) because their cross-sectional distributions vary in shape across time periods; within any single month, the standard deviation is 1.0 by construction.

Feature	**Pooled Mean**	**Pooled Std**
revenue	-0.057	0.681
net_income	-0.063	0.595
operating_cash_flow	-0.053	0.643
total_assets	-0.059	0.509
ret_21d	-0.020	0.957
ret_63d	-0.007	1.007
ret_126d	-0.017	0.977
ret_252d	0.008	0.869

Table 8: Pooled z-score statistics across all dates. Within any single date, std = 1.0 by construction; pooled values < 1.0 reflect time-varying cross-sectional shape.

D Triple-Barrier Labeling Evidence

Supports: Section 2.3

Configuration: upper/lower barrier multiplier = 2.0× EWMA volatility, vertical barrier = 43 trading days (≈62 calendar days), EWMA span = 50 days.

D.1 Label Distribution by Barrier Type

Barrier Type	Label = −1	Label = 0	Label = 1	Total	%
Upper (take-profit)	0	0	12,503	12,503	16.7%
Lower (stop-loss)	7,577	0	0	7,577	10.1%
Vertical (timeout)	21,945	45	32,807	54,797	73.2%
Total	**29,522**	**45**	**45,310**	**74,877**	**100%**

Table 9: Triple-barrier label distribution. 73.2% hit the vertical barrier (timeout).

D.2 Holding Period Statistics (Calendar Days)

Barrier	Count	Mean	Std	Min	Median	Max
Lower (SL)	7,577	37.0	15.2	1	38	276
Upper (TP)	12,503	39.8	14.6	1	41	67
Vertical	54,797	62.4	1.4	60	62	135

Table 10: Holding periods in calendar days. The vertical barrier is 43 trading days, which translates to ≈62 calendar days. Similar TP/SL durations validate the volatility-scaling calibration.

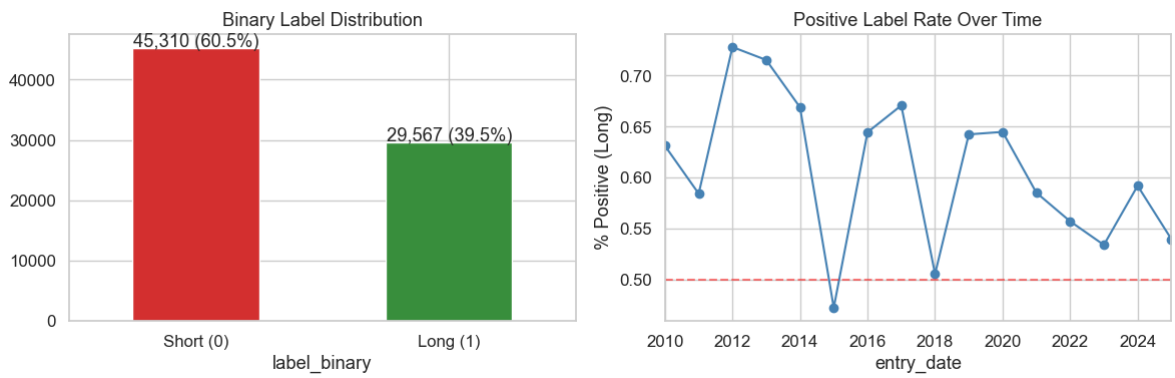

Figure 3: Binary label balance and temporal positive-label rate. The aggregate split (60.5% positive, 39.5% negative) and the visible post-2022 softening are consistent with the train/test balance shift reported in Table E.5.

E Two-Stage Model & CPCV Validation

Supports: Sections 2.4, 2.5, 3.1, 3.2

E.1 Three-Way Temporal Split

E.2 CPCV Summary Statistics (36 Folds)

E.3 Test-Set Classification Metrics

E.4 Probability Calibration

E.5 Label Balance by Split

Segment	Period	Observations	Purpose
Train	2010-03-31 – 2022-11-30	59,533	Fit primary + meta models
Validation	2023-01-31 – 2024-12-31	9,986	Bet sizing grid search
Test	2025-01-31 – 2025-10-31	4,759	Final OOS evaluation

Table 11: Three-way temporal split. The test set was never accessed until final evaluation.

Metric	Count	Mean	Std	Min	25%	50%	75%	Max
Accuracy	36	0.5386	0.0631	0.4088	0.5243	0.5601	0.5854	0.6131
Log Loss	36	0.7519	0.0571	0.6879	0.7182	0.7291	0.7669	0.8773
AUC	36	0.5002	0.0229	0.4454	0.4881	0.5037	0.5125	0.5371

Table 12: CPCV results across 36 folds. Overfitting ratio: 1.14 (max/mean accuracy).

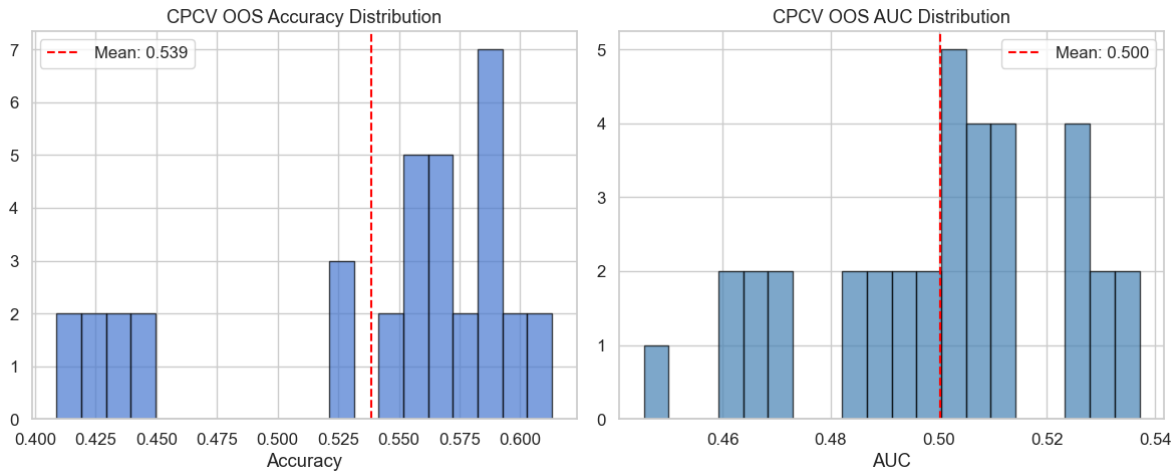

Figure 4: Distribution of CPCV out-of-sample accuracy and AUC across 36 folds. Red dashed lines mark fold means (accuracy ≈0.539, AUC ≈0.500), matching Table E.2.

Model	Accuracy	AUC	Precision (+)	Recall (+)
Stratified Dummy	51.6%	0.500	0.531	0.516
XGBoost (primary)	53.1%	0.502	0.550	0.620

Table 13: Test-set classification: XGBoost vs. stratified dummy baseline. The model's value lies not in ranking power (AUC ≈ 0.50) but in calibrated probabilities that enable effective bet sizing.

Metric	Value
Brier score (model)	0.247
No-skill climatology Brier (predict base rate)	0.249
Expected Calibration Error (10-bin)	0.018

Table 14: Calibration diagnostics. The model improves slightly versus a no-skill climatology baseline on Brier score, while low ECE indicates small average probability-frequency gaps. A reliability diagram is reported in the notebook calibration artifact and should be interpreted jointly with these scalar metrics.

Metric	**Train**	**Test**
Observations	59,533	4,759
Positive (TP hit) %	61.8%	53.1%
Negative (SL hit / timeout) %	38.2%	46.9%

Table 15: The shift from 61.8% to 53.1% positive labels confirms a genuine regime change in 2025, explaining the Sharpe degradation from validation (1.06) to test (0.41).

F Bet Sizing Grid Search

Supports: Sections 2.4, 3.2

F.1 Top Grid Search Configurations (Validation Period)

Rank	Method	Min Prob	Long Leg	Max Cap	Sharpe	Return
1	half_kelly	0.49	top_decile	5%	1.064	55.0%
2	half_kelly	0.49	top_decile	10%	1.064	55.0%
3	half_kelly	0.51	top_decile	5%	1.037	53.8%
4	half_kelly	0.51	top_decile	10%	1.037	53.8%
5	meta_prob	0.51	top_vigintile	10%	0.998	85.8%

Table 16: Top 5 configurations by Sharpe on validation. Row 1 was selected for the final test. The top two configurations are identical at the 5% vs. 10% cap, suggesting robust performance insensitive to the cap choice.

F.2 Grid Search Robustness

Of the 48 configurations tested, 42 produced positive Sharpe ratios on the validation period. The top 10 configurations all used half-Kelly sizing with top-decile selection, varying only in meta-probability threshold (0.49–0.53) and cap level. This consistency across the grid provides evidence that the selected configuration is not an outlier driven by overfitting to one parameter combination.

G Risk Neutralization

G.1 Neutralization Validation Report

Factor	Slope	P-Value	R-Squared	Neutralized
Sector (GICS dummies)	0.000	1.000	0.000	Yes
Size (log market cap)	0.000	1.000	0.000	Yes
Beta (vol_20d proxy)	0.083	0.000	0.004	No

Table 17: Sector and size are fully neutralized. Beta neutralization was attempted but a residual loading persists ($R^2 = 0.004$), representing a small and economically negligible market-beta tilt.

H Backtest & Portfolio Performance

Supports: Sections 1, 3.3, 4

H.1 Full Performance Metrics (Net of Costs)

H.2 Regime-Decomposed Performance

Market regimes were classified using trailing 126-day cumulative S&P 500 returns, binned into bull (>10%), sideways (−10% to 10%), and bear (< −10%) categories. No bear regime occurred during the January–October 2025 test window. Four trading days at regime boundaries are excluded from the decomposition.

Metric	**Value**
Total Return	7.40%
Annualized Return	6.71%
Annualized Volatility	16.50%
Sharpe Ratio	0.41
Sortino Ratio	0.54
Maximum Drawdown	−18.11%
Calmar Ratio	0.37
Daily Hit Rate	51.62%
Monthly Positive %	57.14%
Trading Days	277
Rebalance Dates	8 (monthly)
Avg. Long Positions per Rebalance	≈60 (top decile)

Table 18: Complete out-of-sample performance, net of 7.5 bps/side costs, t+1 staggered execution.

Metric	**Sideways (167 days)**	**Bull (106 days)**
Total Return	7.51%	10.60%
Annualized Return	11.55%	27.07%
Annualized Volatility	17.97%	10.51%
Sharpe Ratio	0.64	2.58
Sortino Ratio	0.98	4.41
Maximum Drawdown	−10.03%	−3.90%

Table 19: Regime-decomposed performance. The strategy captures strong alpha in trending markets and remains profitable in sideways conditions.

I SHAP Explainability & Model Transparency

Supports: Sections 4.2, 4.4

SHAP (SHapley Additive exPlanations) analysis reveals that the model's predictions are driven primarily by **growth** (revenue_growth_yoy, net_income_growth_yoy), **value** (earnings_yield), and **volatility** (vol_60d, gk_vol_20d, mdd_126d) features. Momentum features contribute but are not the dominant driver. This is consistent with the model learning to identify quality/growth stocks with controlled volatility profiles.

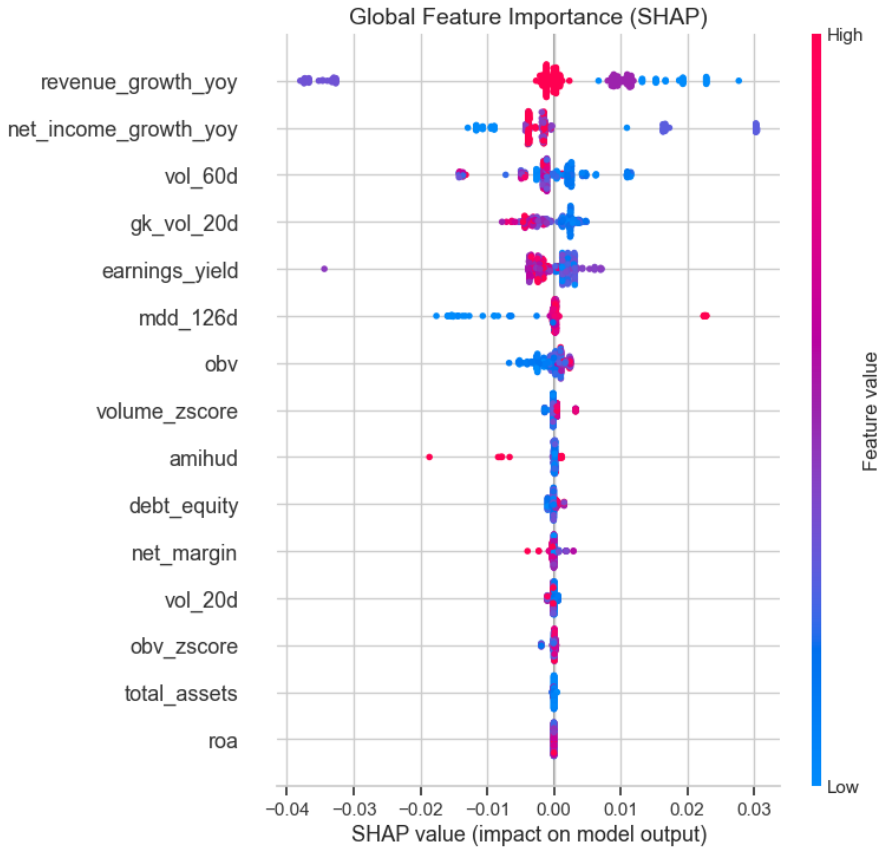

Figure 5: Global SHAP summary plot. The highest-impact features are led by revenue growth, net income growth, earnings yield, and volatility variables, consistent with the narrative in this section.

I.1 PM Decision Ledger (Sample Recommendations)

Rank	Ticker	Sector	Direction	Meta Conviction
1	DUK	Utilities	Long	0.720
2	CMS	Utilities	Long	0.719
3	NI	Utilities	Long	0.719
4	LNT	Utilities	Long	0.719
5	EVRG	Utilities	Long	0.719

Table 20: Top-ranked recommendations from the most recent rebalance. Utilities concentration is mitigated by the 30% sector cap in the portfolio construction layer.

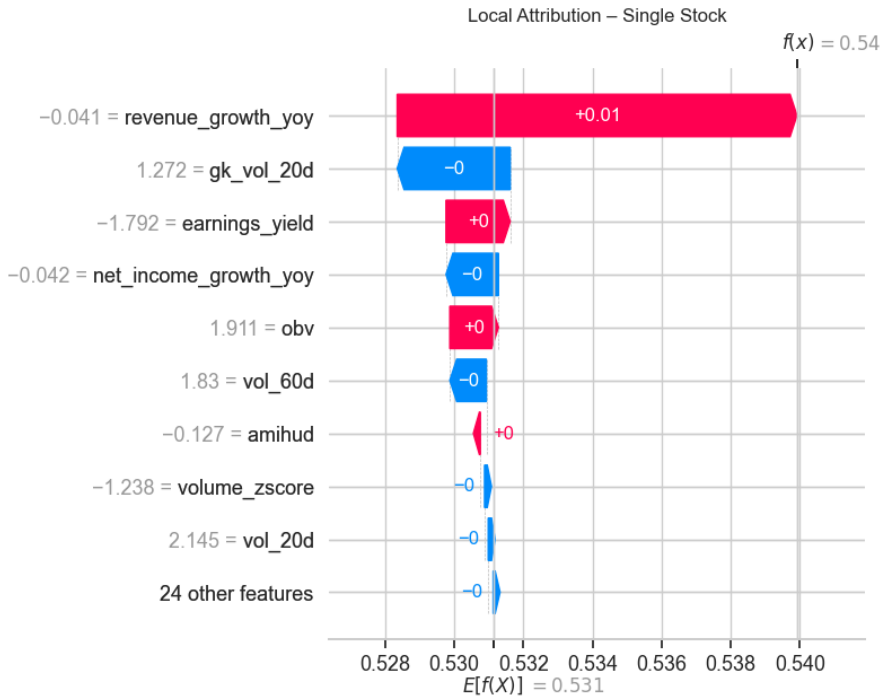

Figure 6: Local SHAP decomposition for a representative single-stock prediction. The plot shows additive feature contributions from the baseline score to the final model output.

J Observation-Level Return Diagnostics

J.1 Scored vs. Allocated Observation Statistics

K Data Integrity Audit

Supports: Sections 1, 3

K.1 Data Leakage Audit Summary

K.2 Key Safeguards Implemented

Metric	Value
Total test observations scored	4,759
Observations with non-zero weight	2,310 (48.5%)
Winning scored observations	2,391 (50.2%)
Losing scored observations	2,368 (49.8%)
Average return (all scored)	+1.32%
Average winner	**+10.45%**
Average loser	**−7.90%**
Profit factor	**1.34**
Best single return	+242.84% (SNDK)
Worst single return	−59.55% (SMCI)
Unique tickers scored	597

Table 21: Observation-level return diagnostics. Counts above separate the full scored universe from the non-zero-weight allocated subset to avoid trade-count ambiguity.

#	Issue	Resolution	Impact
1	Early stopping used test data	Carved internal val from train	Prevents model selection leak
2	Backtest started at t	Shifted to $t+1$ execution	No same-day return capture
3	Neutralization used full window	Per-date cross-sectional only	No future exposure info
4	No 3-way split for sizing	Added validation period	Params unexposed to test

Table 22: Four leakage fixes applied during development.

Safeguard	Description
PIT fundamentals	All filings anchored to SEC acceptance date, not fiscal period end
Survivorship-free	801 tickers including ∼298 historically removed companies
CPCV validation	36 purged folds with 5-day embargo prevent temporal leakage
Three-way split	Train / Validation / Test ensures bet sizing is not optimized on test
t+1 execution	Signals at close of t, execution begins at $t+1$
Transaction costs	7.5 bps/side applied to all reported metrics
Risk neutralization	Sector and size fully orthogonalized; residual beta tilt is negligible ($R^2 = 0.004$)
FRED vintage	Macro layer uses current-vintage FRED; strict PIT vintage reconstruction would require ALFRED/release-aware handling

Table 23: Complete list of pipeline safeguards.